

Scale or Shape? Decomposing NTK Movement During Grokking

Group G2

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Notes for the group. Delete this block before submission.

- [TODO: Later, replace every mention of “rotation” with “shape”, including R_t ’s name, the figure labels and the tier titles. The word “rotation” is kept for now so that the text matches what the group already uses.]
- This is a skeleton. Each section opens with the claim it has to support. Red numbers with an item ID are planning values from read-only checks on `data/` (5 October 2026). Recompute every one through its item in `EXPERIMENTS_TODO.md` before it becomes final.
- The main text must end by page 12. References and appendices do not count, and the appendices may take at most 5 pages. Each section shows its page budget.
- The draft notes after the consent page hold the criteria self-assessment, the changelog of what was cut and why, and the reference checks.
- Each section has a red block headed “Draft contribution”. It says what the section adds and why it matters, written as if the preliminary findings hold. The blocks carry the narrative and answer the marking criteria for significance and novelty. Check each against the full-run results, fold what survives into the section’s prose, and delete the block. Until then the blocks push the main text past its page budget.

Abstract

[TODO: Write last, from the finished sections. Keep one sentence for each point below.] Grokking is delayed generalisation, and what sets the length of the delay is disputed. Recent accounts time it by the weight norm, by the scale of the logits, or by the weight-decay rate. We split the movement of the neural tangent kernel into a change of scale and a rotation, and measure both, with the kernel’s alignment to the labels, on 720 runs of a two-layer network learning modular addition. The runs cross laziness, width and weight decay. At a fixed width, grokking happens at a nearly fixed amount of rotation, but no statistic takes a fixed value across widths. The grokking time depends on both width and weight decay, which neither a rotation clock nor a weight-decay clock predicts alone. Read before any run groks, rotation is the best single predictor of the grokking time, and scale adds information that rotation lacks. [TODO: Final sentence: what this means for the accounts of grokking.]

1 Introduction

Page budget: 1.0 page.

Grokking is the delayed generalisation that Power et al. (2022) found on small algorithmic datasets. A network fits its training data early, and its test accuracy rises from chance long afterwards. Modular addition learned by a two-layer perceptron is a small instance of it. Full-batch gradient descent on the squared error groks there with no explicit regularisation (Gromov, 2023). What sets the length of the delay is an open question.

One account holds that grokking is a transition from lazy training to rich training (Kumar et al., 2024). In lazy training the network behaves as a kernel method with its initial tangent kernel. In rich training the kernel adapts to the task. The evidence that the kernel changes is a distance: the empirical tangent kernel stays close to its initial value until the training data are fitted, and moves far from it afterwards (Mohamadi et al., 2024). A distance does not say

how the kernel changed. A kernel can grow in scale while its eigenvectors stay fixed. It can also rotate, so that its eigenvectors change.

The difference matters for the accounts of the grokking time. Recent work times grokking by the weight norm (Khanh et al., 2026), by the scale of the logits (Khanh, 2026), or by the weight-decay rate (Kim, 2026; Pracher et al., 2026). The result that Kumar et al. (2024) cite for weight decay forcing the kernel to change is that of Lewkowycz and Gur-Ari (2021), in which the kernel only shrinks. None of these accounts measures whether the kernel rotates.

We ask what sets the grokking time: the shape of the kernel, its scale, or the weight-decay rate. Our hypothesis is that rotation sets it. We measure three statistics of the empirical tangent kernel along training: its scale S_t , its rotation R_t and its alignment A_t with the labels. We do so on a grid of 720 runs that crosses laziness, width and weight decay. Our contributions are as follows.

1. We split the movement of the kernel into a change of scale and a rotation, and use each one as a timing variable. On the baseline of Kumar et al. (2024), the usual distance is **mostly a change of scale**^[N2].
2. We test whether grokking happens at a fixed value of each statistic. A placebo test reads each run at another run’s grokking step, so a statistic that merely stays in a narrow band does not pass.
3. We cross width with weight decay and fit how the grokking time scales with each. A rotation clock and a weight-decay clock predict different exponents.
4. We test whether the statistics, read before any run groks, predict when it will. Unlike the settings that produced them, these statistics can be measured in any network.
5. **[TODO: A theory contribution, if section 4.5 keeps one. Its form depends on the option the group chooses there.]**

[TODO: One sentence of findings, matching the abstract, once the numbers are final.]

Draft contribution, Introduction, if the preliminary findings hold. *What is new.* The works we cite each follow one quantity: the distance the kernel moves (Mohamadi et al., 2024), the weight norm (Khanh et al., 2026), the logit scale (Khanh, 2026), or the weight-decay rate (Kim, 2026; Pracher et al., 2026). None of them separates the growth of the tangent kernel from its rotation. None tests a scale clock, a rotation clock and a weight-decay clock against each other on one grid. This report does both, and it adds a placebo test that any claim of a fixed value at grokking can be given. *Why it matters.* The 2026 accounts disagree about what times grokking. If the preliminary findings hold, each account is partly right. Weight decay shortens the grokking time through the scale of the kernel, width lengthens it through the rotation, and neither sets the time alone. The findings also qualify the main evidence for the lazy-to-rich account. At the baseline most of the kernel’s movement is growth^[N2], so a large distance does not by itself show that features were learned. Last, the statistics can be read from any network during training, without knowing its laziness or its effective regularisation. Rotation read before any run groks already orders runs by their grokking time (concordance 0.71 on held-out cells^[E10]). **[TODO: Search the literature beyond the works cited before calling any of this the first of its kind.]** *Verify with the full-run experiments before keeping any of this. If a finding it rests on does not hold, rewrite or drop the claim.*

2 Related Work

Page budget: 1.2 pages.

Grokking. Power et al. (2022) report grokking on small algorithmic datasets. Nanda et al. (2023) reverse-engineer the algorithm a small transformer learns for modular addition and split training into memorisation, circuit formation and clean-up. Gromov (2023) gives closed-form weights that solve modular addition with a quadratic activation, and observes grokking under full-batch gradient descent with no explicit regularisation.

Lazy and rich training. In the infinite-width limit the tangent kernel stays constant, so the network trains as a kernel method (Jacot, Gabriel, and Hongler, 2020). Chizat, Oyallon, and Bach (2020) show that lazy training is a matter of scale. Multiplying a model’s output by a factor α brings its training closer to that of its linearisation as α grows. Kumar et al. (2024) use this factor to move a network between the two regimes, and propose that grokking is the transition from lazy to rich dynamics. Mohamadi et al. (2024) prove that a permutation-equivariant kernel method needs a constant fraction of the p^2 pairs to learn modular addition, and observe that the kernel moves far from its initial value only after the training data are fitted.

What sets the grokking time. Khanh et al. (2026) find that networks trained with weight decay grok when the weight norm reaches a critical value, and that clamping the norm above that value delays grokking exponentially. Khanh (2026) show that under cross-entropy the norm acts through the scale of the logits, and that under squared error it acts through a different route. Kim (2026, Cor. 5, Eq. 20, arXiv v1) derives, for linear models trained with heavy-ball momentum β and weight decay, a grokking time of about $(1 - \beta)/(\eta\lambda)$ times a logarithm of an initial amplitude, and observes this scaling on modular addition. Pracher et al. (2026) predict, for homogeneous networks trained on the squared error with weight decay, that the grokking time is set by the product of the learning rate and the weight decay.

The kernel under weight decay. For a network trained by gradient flow with L_2 strength λ , the kernel evolves as

$$\begin{aligned}\frac{d\Theta_t(x, x')}{dt} &= -2(k-1)\lambda \Theta_t(x, x') + T_t(x, x') + T_t(x', x), \\ T_t(x, x') &= - \sum_{x''} \partial_{\mu\nu} f_t(x) \partial_{\mu} f_t(x') \partial_{\nu} f_t(x'') \ell'(x''),\end{aligned}\tag{1}$$

where k is the degree of homogeneity of the network, the sum runs over the training inputs x'' and repeated parameter indices, and ℓ' is the derivative of the loss with respect to the output (Lewkowycz and Gur-Ari, 2021, Eqs. S5 and S6, arXiv v2). The weight decay appears only in the first term, which rescales the kernel. Only T_t can rotate it. At initialisation T_0 has expectation of order $1/N$ in the width N , a result that Lewkowycz and Gur-Ari (2021) take from Dyer and Gur-Ari (2020). In the infinite-width limit the kernel therefore only decays, with fixed eigenvectors (Lewkowycz and Gur-Ari, 2021, Thms. 1 and 2). **[TODO: Dyer and Gur-Ari (2020) is named here only as Lewkowycz and Gur-Ari’s source. Add it to ref.bib only after checking it on arXiv.]**

Predictions. Table 1 sets out what each account predicts for the three tests of this report. If rotation sets the clock and T_t stays of order $1/N$ along training, a wider network rotates more slowly and groks later at a fixed weight decay. The weight decay should then matter little. A weight-decay clock predicts the opposite.

Table 1: What each account predicts for the three tests. The exponents are those of $\log T = c + a \log N + b \log(\eta\lambda)$. “Not stated” means that the source makes no claim. Entries marked “ours” are our reading of the source.

Account	Fixed at grokking (Tier 1)	Width exponent a (Tier 2)	Decay exponent b (Tier 2)	Early predictor (Tier 3')
Rotation clock (our hypothesis)	R_t reaches a threshold	$a > 0$, since T_t shrinks with width (ours)	$b \approx 0$	R_t
Weight-decay clock (Kim, 2026; Pracher et al., 2026)	Not stated	$a \approx 0$; width can enter Kim’s law only inside the logarithm (ours)	$b \approx -1$	$\eta\lambda$ alone
Norm or scale clock (Khanh, 2026; Khanh et al., 2026)	The weight norm reaches a critical value; in our terms S_t at grokking is fixed (ours)	Not stated	Not stated	S_t

Draft contribution, Related Work, if the preliminary findings hold. *The gap.* Table 1 shows that the three accounts make different predictions for the same experiments, yet each was tested on its own. The accounts that time grokking by weight decay also leave open what sets the time without it. Kim (2026, p. 3, arXiv v1) lists grokking without explicit weight decay, and the creation of new directions by feature learning, among the things its mechanism does not explain. Rotation of the tangent kernel measures one form of that feature learning (our reading). *Why it matters.* A test that sets the accounts against each other on one grid shows which part of each one holds. *Verify with the full-run experiments before keeping any of this.* *If a finding it rests on does not hold, rewrite or drop the claim.*

3 Methods

Page budget: 1.8 pages.

3.1 Task, network and training

The task is modular addition with modulus $p = 23$, at the baseline of Kumar et al. (2024, App. 8.3, arXiv v3). Each pair (a, b) of residues is encoded as the concatenation $x_{a,b} \in \mathbb{R}^{2p}$ of the one-hot codes of a and b . Its target $y_{a,b} \in \mathbb{R}^p$ is the one-hot code of $(a + b) \bmod p$. The dataset holds all $p^2 = 529$ pairs. Each seed draws its own split, with 476 pairs for training and 53 for testing, and its own initial weights.

The network is a two-layer perceptron without biases,

$$f(x; \theta) = N^{-1/2} W^{(2)} \phi(D^{-1/2} W^{(1)} x), \quad (2)$$

with input dimension $D = 2p$, width N , weights $W^{(1)} \in \mathbb{R}^{N \times D}$ and $W^{(2)} \in \mathbb{R}^{p \times N}$ drawn from $\mathcal{N}(0, 1)$, and the rectifier $\phi(h) = \max\{h, 0\}$. The factor $N^{-1/2}$ is the NTK parameterisation of Jacot, Gabriel, and Hongler (2020). Following Kumar et al. (2024), the trained predictor is $\tilde{f}(x; \theta) = \alpha[f(x; \theta) - f(x; \theta_0)]$ with learning rate $\eta = \eta_0/\alpha^2$. A larger α gives lazier training. The loss is the squared error averaged over the outputs and the training pairs. Training is full-batch

gradient descent with weight decay, $\theta_{k+1} = (1 - \eta\lambda)\theta_k - \eta\nabla L(\theta_k)$. For plain gradient descent this is the same as L_2 regularisation (Kim, 2026). Table 2 lists every setting.

Table 2: Every setting of the runs. One setting of $(\alpha, N, \eta\lambda)$ is a cell, and each cell has 5 seeds.

Setting	Value
Modulus, pairs	$p = 23$, all 529 pairs; 476 for training and 53 for testing
Split and seeds	Model seed $s \in \{0, \dots, 4\}$; the split uses data seed $42 + s$
Network	Equation. (2), no biases, rectifier, weights from $\mathcal{N}(0, 1)$
Width N	$\{50, 100, 200, 400, 800, 1600\}$
Output scale α	$\{0.5, 1, 2\}$
Learning rate	$\eta = \eta_0/\alpha^2$ with $\eta_0 = 100$
Weight decay $\eta\lambda$	$\{0, 3 \times 10^{-6}, 10^{-5}, 3 \times 10^{-5}, 10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}\}$
Optimiser and loss	Full-batch gradient descent, no momentum; mean squared error
Budget	200 000 steps
Checkpoints	307 per run: step 0, 120 log-spaced steps to 200 000, and every 1 000 steps
Saved weights	48 steps per run
Runs and compute	720 runs; 68 056 s of wall-clock time ^[N1]

3.2 The grid

Each axis of the grid tests one account. The output scale α sets how lazy the training is (Chizat, Oyallon, and Bach, 2020). The width N sets the size of T_t in (1), and so how fast the kernel can rotate. The weight decay $\eta\lambda$ sets the clock of Kim (2026). The grid has $3 \times 6 \times 8 = 144$ cells and 720 runs, and every run finished.

3.3 Scale, rotation and alignment

We measure the tangent kernel of one scalar output, the sum of the logits divided by $\sqrt{p}$, $g(x; \theta) = p^{-1/2} \sum_c f_c(x; \theta)$. This is the pseudo-NTK of Mohamadi, Bae, and Sutherland (2022). On the 53 test pairs of a seed, the kernel at step t is the matrix with entries $[K_t]_{ij} = \langle \nabla_{\theta} g(x_i; \theta_t), \nabla_{\theta} g(x_j; \theta_t) \rangle$. The kernel of $\tilde{f}$ is α^2 times the kernel of f , so the ratios below do not depend on which of the two is used. With Y the one-hot targets of the test pairs and $C = I - \frac{1}{53} \mathbf{1}\mathbf{1}^\top$, we record

$$S_t = \frac{\|K_t\|_F}{\|K_0\|_F}, \quad R_t = 1 - \frac{\langle K_t, K_0 \rangle_F}{\|K_t\|_F \|K_0\|_F}, \quad A_t = \frac{\langle CK_t C, CYY^\top C \rangle_F}{\|CK_t C\|_F \|CYY^\top C\|_F}. \quad (3)$$

The scale S_t is 1 at initialisation, and a value above 1 means the kernel grew. The rotation R_t is the kernel distance of Fort et al. (2020). It is 0 at initialisation and does not change when the kernel is rescaled. The alignment A_t is the centred alignment of Cortes, Mohri, and Rostamizadeh (2024) with the label kernel $YY^\top$, and it also does not change when the kernel is rescaled.

The usual measure of kernel movement is the relative variation $D_t = \|K_t - K_0\|_F / \|K_0\|_F$ (Geiger et al., 2020). It is a function of the other two,

$$D_t^2 = S_t^2 - 2S_t(1 - R_t) + 1, \quad (4)$$

(appendix A). A kernel that only grows has $R_t = 0$ and $D_t = |S_t - 1|$. A large D_t therefore does not show that the kernel rotated.

[TODO: Add a notation table here if the group wants one: $S_t, R_t, A_t, T, t^*, a, b$, with the value of each at initialisation and how to read it. It costs about a quarter of a page.]

3.4 Event times

The memorisation step of a run is the first checkpoint at which its training accuracy reaches 0.99. Its grokking step T is the first checkpoint at which its test accuracy reaches 0.95. Appendix F repeats the analysis at the levels 0.80 and 0.90. Both events are counted from step 0. A run that does not reach a level within 200 000 steps is kept, and its grokking step is recorded as censored at the budget. The events are read on the checkpoint grid, so each one is known to lie between two checkpoints (at most a tenth of the event step at $\alpha = 1$)^[N3]. We use accuracy rather than loss because the test loss levels off above zero after the test accuracy is perfect (between 0.017 and 0.029 in the baseline cell)^[N4].

3.5 Analysis

Thresholds (Tier 1). For each statistic we take its value at the grokking step of every run that groks, and measure the spread across runs by the coefficient of variation (CV). A statistic can have a small spread because it stays in a narrow band all through training. The placebo test separates the two cases. It reads each run at the grokking step of another run, drawn by a random permutation with no fixed points, and measures the CV again. The tightening ratio is the median placebo CV over 2 000 permutations divided by the CV at the true events. A ratio near 1 means the statistic is not tied to the event.

Scaling (Tier 2). We fit the log-normal accelerated failure time model

$$\log T = c + a \log N + b \log(\eta\lambda) + \gamma \log N \log(\eta\lambda) + \sigma\varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1), \quad (5)$$

by maximum likelihood. A censored run contributes the probability that its grokking step exceeds the budget (Kalbfleisch and Prentice, 2002). Both logarithms are centred at the middle of the grid, so a and b are the slopes there. The fits use lifelines (Davidson-Pilon, 2019). The intervals resample [TODO: whole cells, and seeds within cells, so that they include the misfit of the cells about the fitted plane].

Prediction (Tier 3'). We read S_t , $\log R_t$ and A_t at $t^* = 2430$, a saved step before any run reaches the grokking level. Four model families map them to the grokking step: kernel ridge regression with a Gaussian kernel, a linear accelerated failure time model with interaction terms, gradient-boosted trees with the accelerated failure time objective (Chen and Guestrin, 2016), and a gated recurrent unit (Cho et al., 2014) that reads the statistics at the 25 saved steps up to t^* . Two scores are used. R^2 is the share of the spread in $\log T$ explained on held-out runs. The concordance index is the share of pairs of runs that a model puts in the right order, so that 0.5 is a coin toss (Harrell et al., 1982). The five seeds of a cell are close copies, so whole groups are held out: one cell at a time, which tests interpolation, or one value of α at a time, which tests extrapolation. Appendix E gives the settings of each model.

4 Experiments

4.1 Baseline: the reproduction of Kumar et al.

Page budget: 0.8 page.

Claim. At the baseline of Kumar et al. (2024) the network memorises within a few thousand steps and groks much later, and a lazier network groks later. Over this time the kernel grows several-fold while it rotates by a few per cent.

Figure to make. E1. Four panels against log step: (a) training and test accuracy, (b) S_t , (c) R_t , (d) A_t . Cells with $N = 100$, $\eta\lambda = 0$ and $\alpha \in \{0.5, 1, 2\}$, one colour per α . Median over 5 seeds with the range shaded. Hollow marker at memorisation, filled marker at grokking.

Figure 1: Training at the baseline of Kumar et al. (2024) for three values of α . **[TODO: Final caption.]**

[TODO: One paragraph. Give the median memorisation and grokking steps for each α , with their ranges and the number of seeds that grok within the budget (N5). Say that grokking comes later as α grows, as in Kumar et al. (2024), and point to Appendix B for the comparison setting by setting.]

[TODO: One paragraph on the split of D_t by (4). Give the share of D_t^2 carried by the scale at memorisation, at grokking and at the last step (N2), and the final S_t and R_t . This motivates measuring S_t and R_t separately.]

Draft contribution, Section 4.1, if the preliminary findings hold. *What is new.* The reproduction itself is not new. Its purpose is to confirm that the setup groks as published, which the proposal set as the gate for all later work. What is new is the split of the kernel’s movement at the very baseline where the lazy-to-rich account was proposed (Kumar et al., 2024). *Why it matters.* If most of D_t^2 is scale^[N2], the distance that serves as evidence of the transition mostly records growth. A kernel that only grows keeps its eigenvectors, so its features do not change. Rotation has to be measured on its own before a large distance can be read as feature learning. *Verify with the full-run experiments before keeping any of this. If a finding it rests on does not hold, rewrite or drop the claim.*

4.2 Tier 1: is there a threshold?

Page budget: 1.6 pages.

Claim. At a fixed width, grokking happens at a nearly fixed amount of rotation, and the scale at grokking varies much more. The alignment hardly moves during training. Across widths, no statistic takes a fixed value at grokking.

Figure to make. E2. Three panels, one for each of S_t , R_t and A_t at the grokking step. Every run that groks is a point. Left block: $N = 100$, grouped by α and coloured by $\eta\lambda$. Right block: $\alpha = 1$, grouped by N . Each block is labelled with its CV and tightening ratio.

Figure 2: The kernel statistics at the grokking step. **[TODO: Final caption.]**

Table 3: Spread of each statistic across runs that grok, as a coefficient of variation, at the grokking step, at the memorisation step and at another run’s grokking step (placebo). The tightening ratio is the placebo CV over the CV at grokking.

Runs	Statistic	At grokking	At memorisation	Placebo	Tightening
$N = 100$, all α and $\eta\lambda$	S_t	0.48 ^[E3]	[TODO:]	0.62 ^[E3]	1.3 ^[E3]
	R_t	0.20 ^[E3]	[TODO:]	0.37 ^[E3]	1.9 ^[E3]
	A_t	0.08 ^[E3]	[TODO:]	0.09 ^[E3]	1.1 ^[E3]
$\alpha = 1$, all N and $\eta\lambda$	S_t	0.75 ^[E3]	[TODO:]	0.78 ^[E3]	1.0 ^[E3]
	R_t	0.75 ^[E3]	[TODO:]	0.77 ^[E3]	1.0 ^[E3]
	A_t	0.13 ^[E3]	[TODO:]	0.13 ^[E3]	1.0 ^[E3]

At a fixed width, R_t is the statistic most tied to the event: its spread at grokking is **less than half^[E3]** that of S_t , and it widens **1.9-fold^[E3]** when each run is read at another run’s grokking step. The alignment has a small spread, but the placebo test shows that this comes from staying in a narrow band. It moves from **0.398^[N6]** at initialisation to **0.406^[N6]** at grokking (medians over runs). Across widths no statistic passes the placebo test. At grokking, R_t falls from **0.108** at $N = 50$ to 0.019 at $N = 1600$ ^[N7], and S_t falls from 4.1 to 0.58^[N7]. **[TODO: Name the cells that reach the R_t band of the grokking runs without grokking, if any remain on these data (N8).]**

Wider networks grok with less rotation but take longer, which suggests that width sets the rate at which the kernel rotates. Section 4.3 tests this.

Draft contribution, Tier 1, if the preliminary findings hold. *What is new.* This is a threshold test of the tangent kernel at grokking, with a placebo control, across laziness, width and weight decay. The proposal’s hypothesis, that the alignment crosses a threshold shared by every cell, does not hold here in that form. *Why it matters.* Reading each statistic at the true event and at another run’s event separates a value tied to the event from one that stays in a narrow band. Without that check the alignment would have looked like the best threshold (CV 0.08^[E3]). At a fixed width the rotation needed to grok is nearly fixed, so there the grokking time comes down to how fast the kernel rotates. Across widths the needed rotation changes, so no single kernel state marks grokking. Threshold claims of this kind, such as the critical weight norm of Khanh et al. (2026), can be given the same check. *Verify with the full-run experiments before keeping any of this. If a finding it rests on does not hold, rewrite or drop the claim.*

4.3 Tier 2: width or weight decay?

Page budget: 2.0 pages.

Claim. The grokking time depends on both width and weight decay, so neither the rotation clock nor the weight-decay clock of Table 1 holds on its own. Width acts through the rotation of the kernel, and weight decay acts through its scale.

Figure to make. E4. Map over N (rows) and $\eta\lambda$ (columns) at $\alpha = 1$. Colour is the median $\log T$. Each cell is labelled with the number of seeds that grok out of 5, and a letter for cells where no seed groks (M: memorises only, F: forgets, N: never fits).

Figure 3: Grokking time and outcome over width and weight decay at $\alpha = 1$. **[TODO: Final caption.]**

[TODO: One paragraph on Figure 3 (E4). Without weight decay, only $N = 50$ and $N = 100$ grok within the budget. Inside the range where every seed groks, the time falls with $\eta\lambda$ at every width. The largest weight decay at which a width still groks falls as the width grows.]

Figure to make. E5. $\log T$ against $\log(\eta\lambda)$ at $\alpha = 1$, one line per width, seeds as points, censored runs as open markers. Dashed: the fitted plane of (5). Grey reference slopes of 0 (rotation clock) and -1 (weight-decay clock).

Figure 4: How the grokking time scales with weight decay at each width. **[TODO: Final caption.]**

Table 4: Fits of (5) at $\alpha = 1$ with their 95% intervals, beside the predictions of Table 1. The main row uses the cells with $10^{-5} \leq \eta\lambda \leq 10^{-4}$, where every seed at every width groks. The other rows are checks.

Fit	a	b	γ	σ
Rotation clock predicts	> 0	≈ 0		
Weight-decay clock predicts	≈ 0	≈ -1		
Main range, level 0.95	+0.24 ^[E6]	-0.63 ^[E6]	-0.16 ^[E6]	0.38 ^[E6]
Adding $\eta\lambda = 3 \times 10^{-4}$	+0.43 ^[E6]	-0.21 ^[E6]	[TODO:]	[TODO:]
Level 0.80	+0.30 ^[E6]	-0.55 ^[E6]	[TODO:]	[TODO:]
Level 0.90	+0.27 ^[E6]	-0.60 ^[E6]	[TODO:]	[TODO:]
Memorisation step	-1.53 ^[E6]	[TODO:]	[TODO:]	[TODO:]

Both exponents are nonzero. A wider network groks later, as the rotation clock predicts, and more weight decay shortens the time, as the weight-decay clock predicts. Neither exponent takes the value its account predicts. Wider networks memorise sooner, so the width effect on grokking is not a delay in fitting the training data. **[TODO: Interpret the sign of γ once E6 is final: a negative γ means that weight decay matters more at large width.]**

Figure to make. E7. Two maps over N and $\eta\lambda$ at $\alpha = 1$: median $\log R_t$ and median S_t at t^* . Merge into Figure 3 as extra panels if space is short.

Figure 5: Width sets the rotation of the kernel and weight decay sets its scale. **[TODO: Final caption.]**

[TODO: One paragraph on Figure 5 (E7). Narrow networks rotate the most, and weight decay shrinks the kernel. Give the share of the spread of $\log R_t$ and of S_t that each setting explains (N9). This links Tier 2 back to Tier 1: width sets the rotation and weight decay sets the scale, and the grokking time depends on both.]

[TODO: One sentence on the intervals: if they hold the cells fixed, say so, and give the misfit of the cells about the plane beside the spread between seeds.]

Draft contribution, Tier 2, if the preliminary findings hold. *What is new.* This is a crossed grid of width and weight decay, fitted with censoring and set against the predictions of a rotation clock and of the weight-decay clocks of Kim (2026) and Pracher et al. (2026). The grid includes runs without weight decay, which Kim (2026, p. 3, arXiv v1) places outside the scope of the weight-decay mechanism. *Why it matters.* If both exponents stay nonzero and neither takes its predicted value (Table 4), no single published clock is complete. The data then point to two routes that act together. Width acts through the rotation of the kernel, and weight decay acts through its scale (Figure 5). This reconciles the accounts, since each one describes a single route. It also matters in practice. In the NTK parameterisation a wider network groks later at the same weight decay and fails at a smaller weight decay^[E4], so the weight decay that speeds grokking has to be chosen together with the width. *Verify with the full-run experiments*

before keeping any of this. If a finding it rests on does not hold, rewrite or drop the claim.

4.4 Tier 3': predicting the grokking time from the kernel

Page budget: 1.6 pages.

Claim. Read before any run groks, rotation is the best single predictor of the grokking time. Scale adds information that rotation lacks, and alignment adds little. The best models use rotation and scale together and let them interact.

The settings α and $\eta\lambda$ are the controls of this experiment. In a general network the laziness and the effective regularisation cannot be read off directly, while the tangent kernel can be measured. We therefore ask how much of the grokking time the kernel statistics recover on their own. A model that is given the settings serves as a reference that knows more than any network would report.

Table 5: Kernel ridge regression on subsets of the statistics at t^* , for the runs that grok, scored on held-out cells. The last column is the share of the spread of each quantity that lies between cells.

Statistics	R^2	Concordance	Quantity	Between cells
A_t	0.02 [E10]	0.44 [E10]	$\log T$	0.96 [E10]
S_t	0.27 [E10]	0.64 [E10]	S_t	0.99 [E10]
$\log R_t$	0.37 [E10]	0.71 [E10]	$\log R_t$	0.99 [E10]
S_t and $\log R_t$	0.52 [E10]	0.76 [E10]	A_t	0.56 [E10]
$\log R_t$ and A_t	0.39 [E10]	0.73 [E10]		
S_t and A_t	0.26 [E10]	0.65 [E10]		
All three	0.57 [E10]	0.78 [E10]		

Taken alone, $\log R_t$ explains the most of the grokking time (Table 5). Adding S_t raises R^2 from 0.37 to 0.52 [E10], so rotation does not carry all of the information. Adding A_t to both raises it only to 0.57 [E10]. The rotation and the scale interact. Growth of the kernel shortens the grokking time only when the kernel has also rotated (Figure 7; interaction coefficient -0.62 , interval -1.41 to -0.12 [E9]).

Figure to make. E8. Concordance on a held-out α for a ladder of models: each statistic alone, the linear model, the linear model with interactions, kernel ridge, gradient-boosted trees, and the recurrent network on the history of the statistics. A dashed line marks the reference model given the settings. Points show the three held-out values of α , bars their mean.

Figure 6: Predicting the grokking step of runs with an unseen α . [TODO: Final caption.]

[TODO: One paragraph on Figure 6 (E8). On an unseen α each statistic alone ranks runs only a little better than a coin toss (0.52 to 0.55 [E8]). The linear model with interactions reaches 0.62 [E8], gradient-boosted trees 0.71 [E8] and the recurrent

network on the history 0.77^[E8]. The reference model given the settings reaches 0.78^[E8].]

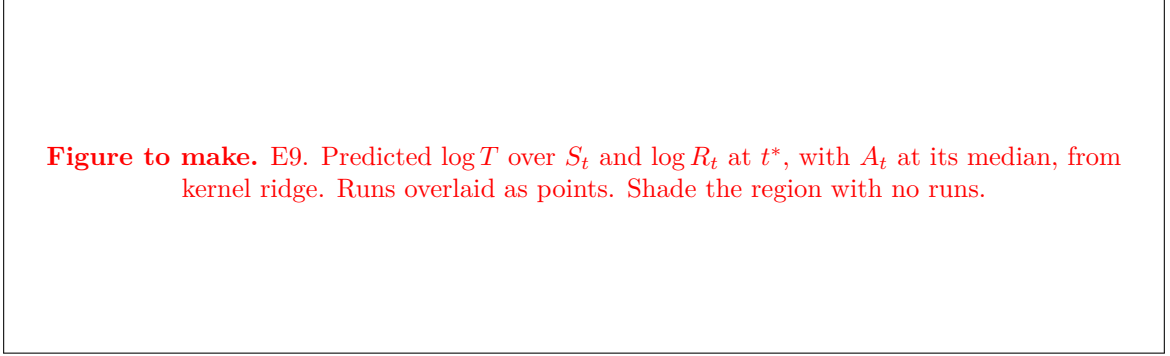


Figure 7: Growth of the kernel shortens the grokking time only when the kernel has also rotated. [TODO: Final caption.]

Draft contribution, Tier 3', if the preliminary findings hold. *What is new.* The works we cite explain the grokking time by settings, such as the weight-decay rate, or by quantities they set directly, such as a clamped weight norm. Here the time is predicted from statistics of the tangent kernel measured before any run groks. The models are scored on cells and on a laziness they have not seen, and each statistic is removed in turn. *Why it matters.* This is a third, separate test of the same answer. Rotation carries the most information, and scale carries information that rotation lacks, so neither one is the whole clock. The statistics can be read from any network without knowing its laziness or its effective regularisation. Early rotation is therefore a candidate progress measure in the sense of Nanda et al. (2023): a quantity that moves before the test accuracy does. Its limits are that the statistics mostly reflect the settings and do not separate the seeds of one cell (section 5). *Verify with the full-run experiments before keeping any of this. If a finding it rests on does not hold, rewrite or drop the claim.*

4.5 Tier 4' (stretch, to be decided): theory

Page budget: 0.6 page.

[TODO: Decide whether this section is worth its 0.6 page, and which theory direction it takes. Decide after E3, E6 and E7 are final, because the right direction depends on whether the mixed clock holds. Record the decision in the changelog, then replace this block with the section.]

Option A: the rate-addition model (the earlier draft). If a width route and a weight-decay route act side by side, their rates add,

$$\frac{1}{T} = r_0 N^{-q} + c \eta \lambda, \quad (6)$$

and the faster route sets the time. Item E11 fits it with censoring and compares it with (5). *Cost:* about a day on existing data. *Value:* a compact form of the mixed clock that may explain the sign of γ . *Risk:* it rests on three assumptions of our own. They are that T_t stays of order $1/N$ along training, that the two routes add, and the form N^{-q} . No source derives the model, and a good fit would not prove the mechanism.

Option B: test the premise behind Table 1. Two checks with no free parameters, on existing data (item E17). The first asks whether the early rate of rotation falls as $1/N$, as the order of T_0 suggests. The second uses the homogeneity of the network. Scaling every weight of (2) by c scales its tangent kernel by c^2 . Weight decay alone, with no gradient, would therefore give $S_t = (1 - \eta\lambda)^{2t}$ and $R_t = 0$. This is our derivation, and it matches the term $-2(k-1)\lambda\Theta_t$ of (1) with $k = 2$. Comparing the measured S_t with this curve shows how much of the change of scale the weight decay accounts for. *Cost*: low. *Value*: it tests the premise that options A and C rest on, against a checked source. *Risk*: low. It is a check, and it is not a new theory.

Option C: can the rotated kernel alone generalise? Freeze the tangent kernel at step t and use it for kernel ridge regression from the training pairs to the test pairs, with a ridge proportional to the kernel’s trace (item E18). Rescaling the kernel then leaves the predictions unchanged, so what the frozen kernel can learn depends only on its shape. Suppose the frozen kernel starts to generalise near the step at which the network groks. Then rotation sets what can be learned, and scale sets only how fast training gets there. That would give a mechanism for the mixed clock. *Cost*: medium. The kernels on all p^2 pairs must be rebuilt from the 48 saved weight steps, which lie up to 20 000 steps apart late in training. *Value*: the highest of the options, since it ties rotation to generalisation directly. *Risk*: medium. A first check is that the frozen initial kernel does not generalise. If it does, the lazy phase alone could generalise, which would challenge the lazy-to-rich reading in this setting.

Option D: the closed-form network. This was the theory tier of the proposal. It bounds $\|T_t\|$ for the two-layer network with a quadratic activation, whose grokked weights are known in closed form (Gromov, 2023). *Cost and risk*: high. It needs new runs with a different activation and new derivations before 9 November.

Option E: drop the section. Spend the space on the Tier 2 checks at $\alpha = 0.5$ and $\alpha = 2$, or move option B to an appendix.

Suggested order. Run option B first. It is cheap, and the other options rest on its premise. If time allows, option C adds the most. Option A is a short add-on. It summarises the data and does not explain them. Option D is likely out of reach.

Draft contribution, Tier 4’, if option C is chosen, if the preliminary findings hold.

What is new. A direct test of whether the rotated tangent kernel alone can generalise, set against the step at which the network groks. *Why it matters.* It would explain the mixed clock found in Tiers 1 to 3’. The shape of the kernel decides what can be learned, and its scale decides how fast training gets there. *Verify with the full-run experiments before keeping any of this.* *If a finding it rests on does not hold, rewrite or drop the claim.*

Figure to make. The figure of the chosen option: E11 for option A ($1/T$ against $\eta\lambda$ with the fitted lines of (6)), E17 for option B (early rotation rate against N , and S_t against the curve of weight decay alone), or E18 for option C (test accuracy of the frozen kernel against step, with the network’s grokking step marked).

Figure 8: **[TODO: Caption for the chosen option.]**

5 Limitations

Page budget: 0.6 page.

[TODO: Write each point as one or two sentences.]

- Every run uses the NTK parameterisation, one task and one architecture. A width effect measured here belongs to this parameterisation.
- Each cell has 5 seeds.
- The events are known only to the resolution of the checkpoint grid.
- The fits in Tier 2 depend on which cells enter, and the decay edge bends the plane (Table 4).
- The kernel statistics say nothing about which seed of a cell groks first, and the reference model given the settings predicts better than any kernel model.
- The alignment here uses the kernel of the summed logits. The tangent kernel of all p logits may behave differently.
- The centred predictor is not homogeneous, so the theorem behind (1) applies to our network only as a guide.
- The proposal also planned a clamped-norm arm, a sweep over p and the training fraction, and a cross-entropy arm. They were not run.
- The proposal defined the events by the loss. The report uses accuracy (section 3.4).

6 Conclusion

Page budget: 0.3 page.

[TODO: Three to five sentences. Answer the research question: rotation matters most, but the grokking time is set by rotation and scale together, through width and weight decay. Restate what is new: the scale and rotation split as timing variables, the placebo threshold test, the crossed grid, and prediction from quantities that any network exposes. Name one next step.]

Draft contribution, Conclusion, if the preliminary findings hold. We asked what sets the grokking time. If the full runs confirm the preliminary findings, rotation of the tangent kernel matters most, but it is not enough on its own. Grokking happens at a nearly fixed rotation only within one width. The grokking time depends on both width and weight decay. Early rotation and scale together predict it better than either one does alone. Each published clock therefore captures one route: weight decay acts through the scale of the kernel, and width

acts through its rotation. What is new is the split of the kernel’s movement into scale and rotation as timing variables, the placebo test of thresholds, the crossed grid of width and weight decay, and prediction from quantities that any network exposes. The next step is **[TODO: the theory direction chosen in section 4.5, or a second task]**. *Verify with the full-run experiments before keeping any of this. If a finding it rests on does not hold, rewrite or drop the claim.*

Contribution

[TODO: One line for each member, from the group.]

- Al-Saffi: **[TODO:]**
- Chong: **[TODO:]**
- D’Mello: **[TODO:]**
- Kienzle: **[TODO:]**
- Rieger: **[TODO:]**

Acknowledgement

[TODO: Anyone outside the group who advised on the project.]

Use of AI tools

[TODO: The task sheet asks each deliverable to state how AI tools were used. The group should agree on this text. A draft follows.] The group used an AI assistant (Claude, by Anthropic) to draft parts of the code, to restructure this report, and to check references against their arXiv pages. Group members checked every result and every reference before submission.

A The identity for the variation

Page budget: 0.5 page (appendices: 5 pages in all).

Expanding the squared norm gives $\|K_t - K_0\|_F^2 = \|K_t\|_F^2 - 2\langle K_t, K_0 \rangle_F + \|K_0\|_F^2$. Dividing by $\|K_0\|_F^2$ and using $\langle K_t, K_0 \rangle_F = (1 - R_t)\|K_t\|_F\|K_0\|_F$ from (3) gives $D_t^2 = S_t^2 - 2S_t(1 - R_t) + 1$. By the Cauchy-Schwarz inequality, $R_t = 0$ exactly when $K_t = cK_0$ for some $c > 0$, and then $D_t = |S_t - 1|$. **[TODO: Check against the golden report, Appendix A.1.]**

B The baseline against Kumar et al.

Page budget: 0.75 page.

[TODO: A table that sets each setting of the baseline beside Kumar et al. (2024, App. 8.3, arXiv v3): task, encoding, training fraction, width, parameterisation, loss, optimiser, learning rate and α values. Open the paper and check every entry (CLAUDE.md, Rule 3). Kumar et al. (2024) also sweep $\alpha = 1.5$, which our grid leaves out. Add the grokking steps by α beside their reported values (E12).]

C Outcomes over the whole grid

Page budget: 1.0 page.

[**TODO:** One table for each α : the seeds that grok out of 5 and the median grokking step for every cell of N and $\eta\lambda$ (E13).]

D A cross-check from a second pipeline

Page budget: 0.75 page.

[**TODO:** One paragraph and one figure. A second training pipeline (release grid-v0.1.0 on branch feat/grid) ran 12 seeds per cell at the baseline with weight decays near the edge, with a budget of 100 000 steps. The golden report gives these results: the grokking time is flat in $\eta\lambda$ below a break at $\eta\lambda \approx 7.0 \times 10^{-4}$ [E14], rises steeply above it, and no seed groks at 8.28×10^{-4} [E14]. Export the figure as a PDF from that pipeline (E14).]

E Settings of the prediction models

Page budget: 1.0 page.

[**TODO:** Describe the grouped cross-validation, the search grid of each model, the architecture and training of the recurrent network, and the random seeds (E15). Notebooks 3 and 4 in `model_fitting/` hold these settings.]

F Sensitivity to the grokking level

Page budget: 0.5 page.

[**TODO:** Repeat Table 3 and Table 4 at the levels 0.80 and 0.90 (E16).]

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Indicate whether all your group members consent for your report to be used as a teaching resource by including one of the statement below.

We give consent for this to be used as a teaching resource.
We DO NOT give consent for this to be used as a teaching resource.

[TODO: Keep one of the two statements.]

Draft notes: delete before submission

1. How the report meets each marking criterion

The task sheet marks the report on five criteria and asks it to cover six points. The table gives where this skeleton addresses each one, the mark we expect if the plan is carried out, and how to raise it.

Criterion	Where it is addressed	Likely mark	How to improve
Clarity and conciseness	One research question. Three statistics on one kernel. One regression model and one cross-validation scheme. Each section opens with its claim, and the spectral and survival machinery is cut.	Good once written. The risk is a heavy load of notation and 8 figures in 12 pages.	Add the notation table. Merge Figure 5 into Figure 3. Use one colour for each statistic in every figure. Do a pass for the writing rules of CLAUDE.md and for spelling.
Correctness	Censored fits, the placebo test, grouped cross-validation, and the caveats stated in section 5.	Good, with a risk in Tier 2.	Report the check rows of Table 4. Make the intervals resample cells. State the assumptions of the theorem behind (1). Label any Tier 4' model that no source derives as the group's own.
Reproducibility	Table 2, the grid, the split rule, the compute, and one script for each figure, listed in the experiments file.	Good.	Publish a small extract of the data. Generate every number as a macro. Add a README that maps each figure to its script.
Significance	Three 2026 accounts tested on one grid, with statistics that any network exposes. The “why it matters” half of each draft contribution block (Sections 1, 2, 4.1 to 4.4 and 6) states the case section by section.	Medium to good.	Keep the reconciling claim (each clock describes one route) only if Tier 2 and E7 confirm it. Say what a practitioner can measure and why it helps. Discuss what a single small task can and cannot show.
Novelty	The five contributions listed in the Introduction. The “what is new” half of each draft contribution block says what each section adds over the works cited.	Medium to good. The split into scale and rotation may look small.	Search the literature beyond the works cited before claiming any first. Contrast the split with the distance of Mohamadi et al. (2024) and with the untested argument of Kumar et al. (2024). Use the scope of Kim (2026), which leaves out grokking without weight decay. Repeat what is new in the Conclusion.
Required content	Problem and significance (Sections 1 and 2), steps (Section 3), findings (Section 4), limitations (Section 5), conclusion (Section 6), contribution, AI use and consent.	Complete.	Check the limits of 12 pages and 5 appendix pages, and name the file G2_report_<Name>.pdf.

2. Changelog: decisions and what was cut

Decisions taken on 5 October 2026.

- All results come from `data/`: 720 runs, 5 seeds, 200 000 steps, $\alpha \in \{0.5, 1, 2\}$, six widths

and eight weight decays. The golden report used the grid-v0.1.0 databases (12 seeds, 100 000 steps) instead.

- The alignment A_t is the centred alignment of the summed-logit kernel on the test pairs (`A_sum` in `data/`). The golden report on main uses the tangent kernel of all p logits, after Baratin et al. The statistics S_t , R_t and A_t now share one kernel, and the prediction notebooks already use this A_t .
- The main text uses one grokking level, 0.95. The levels 0.80 and 0.90 move to Appendix F.
- The reproduction of Kumar et al. (2024) keeps a short section with one figure. The comparison setting by setting moves to Appendix B.
- Tier 4' was first drafted as the rate-addition model. It was reopened the same day as a decision between five options (section 4.5), because that model rests on assumptions of our own.
- Each section gained a red draft contribution block that states, as if the preliminary findings hold, what is new and why it matters.
- The word “rotation” is kept for now and will become “shape” later.
- In Tier 3' the model given the settings is a reference, since a general network does not report its laziness or its effective regularisation.
- The planning checks changed one claim. The alignment does not pass the placebo test, so Tier 1 claims a threshold for rotation only.

Cut from the golden report. Each item can be restored from the golden report’s source, `docs/report_golden.tex` (added in commit 7da1996), and its figure files in `docs/figures/`.

- Section 3.4, Spectral measures: effective rank, trace ratios and the target power along eigenvectors. Section 4.3: the cumulative target power figure (`docs/figures/spectrum.tex`).
- Section 3.5, Event times: the Kaplan-Meier estimate with Greenwood’s interval, the phase labels, the additive model of Aalen and the relative risk model with time-dependent covariates. Section 4.3: the table of that model. Appendix B, Model checks: the probability plot and the cumulative regression functions (`docs/figures/probability.tex`, `aalen.tex`).
- Section 3.6, Choice of levels: optimal designs, the censored information matrix, the broken line and the sequential design. Section 4.2.2: the scan, the design table, the broken-line figure, the survival figure and the accelerated failure time table of the weight-decay arm (`docs/figures/phases.tex`, `broken.tex`, `survival.tex`). A short summary stays in Appendix D.
- Section 4.2.1: the α sweep with $\alpha = 1.5$ and 12 seeds (`docs/figures/alpha_sweep.pdf`).
- Section 3.2: the first-logit kernel as a second output.
- Related Work: the paragraph on kernel alignment and spectral bias, silent alignment, the early and late dichotomy, Omnigrok, and provable grokking in ridge regression.
- The pipeline itself: branch `feat/grid`, folder `grid/` (Snakemake workflow, DuckDB databases, R scripts).

Planned in the proposal and not run. The clamped-norm arm with $\rho \in [1, 2]$, the arm over three primes and three training fractions, the cross-entropy arm, the statistic $y^\top K^{-1}y$, and events defined by loss thresholds. The proposal is `docs/G2_proposal.tex`.

Superseded analyses. `reproduced-code/tier1_v2.ipynb` and `tier2_width_decay.ipynb` use the old data, with one split shared by every seed, the trace kernel and test accuracy 1.0. Their methods are ported, and their numbers are not used.

Open decisions for the group.

- Keep the title, or change it to give the answer.
- Keep Figure 5 as its own figure, or merge it into Figure 3.
- Whether to add the notation table.
- Whether the intervals in Tier 2 should resample cells.
- Which theory direction, if any, Tier 4' takes (options A to E in section 4.5).

3. Reference checks

Every entry of `ref.bib` was checked on 5 October 2026 against its arXiv abstract page, its DOI record on Crossref, or PubMed. Each section and equation number cited in the text was checked in the PDF of the version named.

Entry	Checked against	Status	Notes
Power et al. 2022	arXiv 2201.02177	Matches	v1 only.
Nanda et al. 2023	arXiv 2301.05217	Matches	v3 of 19 Oct 2023 gives the month.
Gromov 2023	arXiv 2301.02679	Matches	
Jacot et al. 2020	arXiv 1806.07572	Matches	Year is v4 (Feb 2020); NeurIPS 2018.
Chizat et al. 2020	arXiv 1812.07956	Matches	Year is v5 (Jan 2020); NeurIPS 2019.
Kumar et al. 2024	arXiv 2310.06110	Matches	v3 of 11 Apr 2024. App. 8.3 still to be opened (Appendix B).
Mohamadi et al. 2024	arXiv 2407.12332	Matches	ICML 2024.
Mohamadi, Bae, Sutherland 2022	arXiv 2206.12543	Matches, entry type changed	The golden bib types it as an article with no journal. It is typed misc here. Published at ICML 2023.
Lewkowycz, Gur-Ari 2021	arXiv 2006.08643, PDFs of v1 and v2	Matches	Year is v2 (Jan 2021); NeurIPS 2020. CLAUDE.md gives both 2020 and 2021. The kernel flow is Eq. S5 and T_t is Eq. S6 in both v1 and v2. The proposal's draft note, which says they become S8 and S9 in v2, is wrong: S8 in v2 is the infinite-width solution. $E[T_0] = O(n^{-1})$ is credited to Dyer and Gur-Ari (2020), which is not in ref.bib. NeurIPS 2020.
Fort et al. 2020	arXiv 2010.15110	Matches	
Geiger et al. 2020	Crossref and arXiv 1906.08034	Matches	JSTAT 2020, 113301.
Cortes et al. 2024	arXiv 1203.0550	Matches	Year is v3 (Apr 2024). The paper is JMLR 13 (2012), 795–828.

Entry	Checked against	Status	Notes
Khanh et al. 2026	arXiv 2606.13753	Matches arXiv; name order flagged	arXiv lists “Truong Xuan Khanh” and parses the surname as Khanh, so “Khanh et al.” matches arXiv. In Vietnamese order the family name is usually written first, which would make it “Truong et al.”. The co-authors have the same question. The proposal’s key is truong2026norm . The group should decide.
Khanh 2026	arXiv 2606.18465	Matches arXiv; name order flagged	As above.
Kim 2026	arXiv 2607.23967	Matches	Cor. 5 and Eq. 20 checked in the v1 PDF (page 8). The Scope paragraph cited in Section 2 and Tier 2 is on page 3 of v1. It lists grokking without explicit weight decay and the creation of new population-visible directions by feature learning among what the mechanism does not explain.
Pracher et al. 2026	arXiv 2609.26679	Matches, fields added	The golden bib gives only the DOI. The eprint, month and class were added. v1 of 22 Sep 2026.
Kalbfleisch, Prentice 2002	Crossref, Open Library	Matches, edition changed	The golden bib gives edition 1. The 2002 Wiley printing is the second edition; the first appeared in 1980. Crossref lists the online version as edition 1, so this needs a check of the book itself.
Davidson-Pilon 2019	Crossref	Matches	JOSS 4(40), 1317.
Harrell et al. 1982 (new)	PubMed 7069920, Crossref	Matches	JAMA 247(18), 2543–2546. Crossref lists only the first author; PubMed lists all five.
Chen, Guestrin 2016 (new)	arXiv 1603.02754	Matches	KDD 2016.
Cho et al. 2014 (new)	arXiv 1406.1078	Matches	EMNLP 2014.